Exhaustive Manual Testing Guide: Settings Sub-Flow (Branch Settings, Financial Years, Dashboard Widgets, States, Cities, Countries)

1. Overview & Architecture Scope

The Settings Sub-Flow provides administrative control over system parameters, global geographic taxonomies (Countries, States, Cities), financial calendar years, branch-level operational configurations, and executive dashboard widget layouts.

Target Components & Resources:

- **Geographic Setup:** Countries, States, Cities
- **Financial Calendar:** Financial Years
- **Operational Configuration:** Branch Settings
- **Dashboard Layout:** Dashboard Widgets

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2. Pre-requisites & Test Environment Setup

1. Laravel/Filament application database migrated cleanly (`php artisan migrate:fresh`).
2. Super Admin user created with full access privileges.
3. At least 2 active physical branches created (`Coimbatore Main Branch`, `Chennai Retail Outlet`).

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3. Comprehensive Test Cases

3.1 Geographic Master Data (Countries, States, Cities)

Test Case 1.1: Country Onboarding & ISO Code Uniqueness (Positive & Negative)

- **Objective:** Verify creation of master country records with strict ISO code validation.
- **Pre-conditions:** Logged in as Super Admin.
- **Action 1 (Negative Validation):** Navigate to `Countries -> Create`. Leave Name empty and enter duplicate ISO code `IN`. Click `Save`.
    - **Expected Result:** System throws validation errors for missing Country Name and unique ISO code violation.
- **Action 2 (Positive Creation):** Enter Country Name `India`, ISO Code `IN`, Phone Code `+91`, Currency Code `INR`. Click `Save`.
    - **Expected Result:** Country created cleanly.

Test Case 1.2: State Management & Statutory Tax Code Linkage

- **Objective:** Verify state entity creation and GST state code mapping.
- **Action:** Navigate to `States -> Create`. Select Country `India`. Enter State Name `Tamil Nadu`, State Code `33`. Create another state `Kerala`, State Code `32`.
- **Expected Result:** States saved. State Code `33` mapped for intra-state tax split validation.

Test Case 1.3: City Creation & Parent State Linkage

- **Objective:** Link cities to parent states.
- **Action:** Navigate to `Cities -> Create`. Select State `Tamil Nadu`. Add Cities `Chennai`, `Coimbatore`, `Madurai`. Select State `Kerala`. Add City `Kochi`.
- **Expected Result:** Cities saved with foreign keys pointing to correct state records.

Test Case 1.4: Cascading Geographic Dropdown Verification in Forms

- **Objective:** Ensure selecting a Country filters States, and selecting a State filters Cities dynamically.
- **Action:** In Branch/Customer/Vendor forms, select Country `India`.
    - **Validation 1:** State dropdown MUST display only Indian states (`Tamil Nadu`, `Kerala`).
- **Action:** Select State `Tamil Nadu`.
    - **Validation 2:** City dropdown MUST display only Tamil Nadu cities (`Chennai`, `Coimbatore`, `Madurai`). Cities from `Kerala` (`Kochi`) MUST be excluded.
- **Expected Result:** Cascading dropdown logic functions without lag or unhandled options.

Test Case 1.5: Foreign Key Deletion Restriction on Geography

- **Objective:** Protect child records from orphan state deletion.
- **Action:** Attempt to delete State `Tamil Nadu` while active Cities (`Chennai`, `Coimbatore`) are linked to it.
- **Expected Result:** System blocks deletion: "Cannot delete state with active linked cities (FK constraint)."

3.2 Financial Years & Accounting Calendar

Test Case 2.1: Financial Year Creation & Valid Date Window

- **Objective:** Define fiscal year calendar windows.
- **Action:** Navigate to `Financial Years -> Create`.
    - Name: `FY 2026-2027`
    - Start Date: `2026-04-01`
    - End Date: `2027-03-31`
    - Status `is_active`: `true`
    - Status `is_closed`: `false`

    - ○ Save.

- **Expected Result:** Saved successfully as active financial year.

Test Case 2.2: Single Active Financial Year Enforcement Rule

- **Objective:** Ensure ONLY ONE financial year can be marked `is_active = true` at any given time.
- **Action:** Create `FY 2027-2028` (Start: `2027-04-01`, End: `2028-03-31`). Set `is_active = true`. Save.
- **Expected Result:** System automatically sets `is_active = false` on `FY 2026-2027` so only `FY 2027-2028` remains active.

Test Case 2.3: Date Overlap Prevention & Validation

- **Objective:** Block creation of overlapping fiscal periods.
- **Action:** Create a financial year with Start Date `2026-10-01` and End Date `2027-09-30`.
- **Expected Result:** System rejects creation with error: "Financial year dates overlap with an existing financial calendar."

Test Case 2.4: Financial Year Closing Workflow (`is_closed = true`) & Transaction Lockout

- **Objective:** Lock past financial years to prevent backdated posting fraud.
- **Action 1 (Close FY):** Select `FY 2025-2026`. Toggle `is_closed = true`. Click `Save`.
- **Action 2 (Lockout Check):** Try to post a POS sale, GRN, or Journal Entry dated `2025-06-15` (within closed FY).
- **Expected Result:** System blocks entry: "Posting rejected. Financial Year FY 2025-2026 is CLOSED."

Test Case 2.5: Cross-Module Financial Year Reference

- **Objective:** Verify Merchandise Planning, Payroll, and TDS modules resolve active FY.
- **Action:** Go to `Merchandise Plannings` / `Employee Tax Declarations`. Click `Create`.
- **Expected Result:** Financial Year field auto-selects the currently active FY by default.

3.3 Branch Settings & Operational Configuration

Test Case 3.1: Branch Setting Initialization & Tenant Linkage

- **Objective:** Configure branch-specific operational parameters.
- **Action:** Navigate to `Branch Settings` -> Select `Coimbatore Main Branch`. Fill:
    - ○ Default Receipt Header: `Apex Textiles – Coimbatore Branch`
    - ○ Receipt Footer: `Thank you for shopping! Exchange within 7 days.`
    - ○ Invoice Prefix: `INV-CBE-`
    - ○ Default Cashier Discount Limit: `15%`
    - ○ Allow Negative Stock: `false`
    - ○ Save.

- **Expected Result:** Branch settings stored per `branch_id`.

Test Case 3.2: Receipt Footer & Header Rendering Verification

- **Objective:** Verify branch settings render on POS receipts.
- **Action:** Complete a POS sale at `Coimbatore Main Branch`. Generate receipt preview.
- **Expected Result:** Receipt header reads `Apex Textiles – Coimbatore Branch` and footer reads `Thank you for shopping! Exchange within 7 days.`

Test Case 3.3: Invoice Prefix & Sequential Counter Formatting

- **Objective:** Ensure invoices follow configured branch prefix.
- **Action:** Generate new B2B Sales Invoice at `Coimbatore Main Branch`.
- **Expected Result:** Invoice Number auto-formats as `INV-CBE-00001`.

Test Case 3.4: Discount Limit & Negative Stock Guardrails

- **Objective:** Enforce branch operational limits.
- **Action 1 (Discount Check):** In POS, cashier attempts 20% discount (limit is 15%).
    - **Expected Result:** Blocked: "Discount exceeds branch limit of 15%."
- **Action 2 (Negative Stock Check):** Attempt sale of out-of-stock item.
    - **Expected Result:** Blocked: "Branch settings prohibit negative stock transactions."

Test Case 3.5: Multi-Branch Setting Isolation

- **Objective:** Ensure Branch A settings do not impact Branch B.
- **Action:** Change `Chennai Retail Outlet` settings (Invoice Prefix: `INV-MAA-`, Discount Limit: `20%`).
- **Expected Result:** `Coimbatore Main Branch` retains `INV-CBE-` and `15%` limit without alteration.

3.4 Dashboard Widgets & Customization

Test Case 4.1: Dashboard Widget Registration & Active Toggle

- **Objective:** Manage dashboard analytics widgets.
- **Action:** Navigate to `Dashboard Widgets` -> Toggle visibility for `Daily Sales Summary`, `Low Stock Alerts`, `Pending Approvals`.
- **Expected Result:** Widget states update in database.

Test Case 4.2: Role-Based Widget Visibility Matrix

- **Objective:** Restrict sensitive widgets based on user role permissions.
- **Action 1 (Admin Login):** Log in as Super Admin.
    - **Expected Result:** Admin dashboard displays ALL widgets (`Daily Sales`, `Gross Profit`, `Payroll Expense`, `Stock Value`).

- **Action 2 (Cashier Login):** Log in as `Branch Cashier`.
    - **Expected Result:** Cashier dashboard displays ONLY `Daily Shift Sales` widget. Financial profit and HR payroll widgets are HIDDEN.

Test Case 4.3: Real-Time Widget Data Calculation Integrity

- **Objective:** Verify math on dashboard summary cards.
- **Action:** Check `Daily Sales Widget` total. Perform new POS sale of `2,500 INR`. Refresh dashboard.
- **Expected Result:** Widget aggregate count increases by exactly `2,500.00 INR`.

Test Case 4.4: Widget Position Drag & Drop Reordering

- **Objective:** Persist user dashboard layout customizations.
- **Action:** Reorder widgets on dashboard (Move `Low Stock Alerts` to position 1). Save Layout. Refresh page.
- **Expected Result:** Layout position preserved after page refresh.

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4. Summary & Pass Criteria

Component	Key Validation	Expected Result	Pass Criteria
Geographic Master	Cascading Selects & ISO Unique	Country -> State -> City cascading works	Pass
Financial Years	Date Overlap & Single Active	Only 1 active FY; Closed FY blocks posting	Pass
Branch Settings	Receipt/Invoice & Guardrails	Branch prefix & discount limits enforced	Pass
Dashboard Widgets	RBAC Visibility & Aggregates	Widgets reflect role permissions & real-time DB math	Pass